Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Decisive
Project / Solution Name	Nexus (NexusTalent)
Name (Leader)	Raye Chan Jun Foong
Email & Phone (Leader)	rayechanjf@gmail.com / 019-8184052
University Affiliation(s)	Universiti Malaya (UM)
Live Prototype URL	https://NexusTalent-Decisive.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/RayeC-1305/XMUMSDC_Decisive

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Nexus uses a privacy-first, decoupled browser-to-API architecture deployed on Replit (https://NexusTalent-Decisive.replit.app) across three boundaries :

1. **Browser Workspace (React 18 / Vite 6 / Tailwind CSS):** Serves both the six-stage individual worker workspace (extracting PDF/DOCX/TXT CVs locally via PDF.js/fflate and saving progress in device localStorage) and the seven-chapter **HR & Employers workflow (/hr, /hr/demo, /survey/:token)**, featuring department radars/heatmaps, 5-second live survey polling, cohort builders, and cited Markdown brief exports .
2. **Career & Workforce API (Node.js 22.16+ / Express 4 CPU Rules Engine):** Processes individual reviewed profiles statelessly in memory and powers HR survey campaigns (seeded with 3 departments × 8 synthetic profiles, expiring/quota-limited /survey/:token links, and a single-process local JSON store for synthetic department data) . Both workflows share the same deterministic matching engine (services/career_engine.js: 45% required skills, 30% transferable bridges, 25% research/scenario context) and authored catalogs (services/career_catalog.js) with zero LLM or GPU runtime dependencies .
3. **Public Research Signal Layer (services/evidence_service.js → Crossref REST API):** Retrieves dated scholarly metadata, caches revision receipts on disk, and applies bounded keyword-to-skill rules with stale-cache fallback .

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript (ES2022+) / HTML5 / CSS3	Core application logic, deterministic scoring engine, UI components, and pitch deck	W3C / ECMA Standard
Node.js (v18+ / v20+)	Backend JavaScript runtime for the Career API and evidence caching service	MIT License
React & React-DOM (v18.x)	Frontend user interface for the 6-stage worker and adviser reskilling workspace	MIT License
Vite (v5.x)	Frontend bundler and local development server	MIT License
Express (v4.x)	REST API server handling `/api/career/*` scoring, pathway, and evidence endpoints	MIT License
Sora, DM Sans, IBM Plex Mono (v17 / v20)	Embedded open-source fonts for UI and self-contained HTML pitch deck	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Crossref REST API (`api.crossref.org` — Crossref)	Retrieves updatable public scholarly publication metadata, DOIs, publishers, and publication dates for AI and workforce trend signals	Free (Public Polite Pool API)
None (No other external runtime APIs)	All role matching, gap analysis, and pathway ordering run locally via deterministic CPU rules	None

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit Inc.)	Cloud hosting, build environment, and live deployment of the full-stack React/Vite frontend and Node.js/Express Career API (`https://NexusTalent-Decisive.replit.app`)	Sponsor-Provided / Replit Cloud (Global)
GitHub (Microsoft)	Public source code repository, version control, and documentation (`RayeC-1305/XMUMSDC_Decisive`)	Free Tier (Global)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None (No active LLM or neural model at runtime)	N/A — Explainable Deterministic CPU Rules Engine (`services/career_engine.js` & `services/evidence_service.js`)	Self-authored explainable rule & weighted index engine (45% required skills, 30% transferable bridges, 25% research/scenario context)	MIT / Team Decisive Original Work

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

4.2 Datasets

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Crossref Public Scholarly Metadata Cache	`https://api.crossref.org/works` (Cached in repo revision `947afd1b22fcbef5`, e.g., DOI `10.1093/haschl/qxag295`)	Public	CC0 1.0 Public Domain	N
Nexus Authored Career, Skill, Scenario & SDG Catalog	`src/data/careerCatalog.js` & `services/career_catalog.js` in repo	Self-collected / Authored	MIT License	N
Curated Public Learning Resource Metadata	Microsoft Learn (`learn.microsoft.com`) & UK Gov Data Quality Framework (`gov.uk`)	Public	Public Web Documentation	N
Fictional Demo Worker Profiles (e.g., Siti Rosli)	`src/data/careerCatalog.js` & `output/pitch/demo-proof.json`	Synthetic	MIT License	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude / OpenAI Coding Assistants	Code scaffolding, CSS layout refinement, documentation formatting, and test script assistance	Moderate
Google Gemini	Reviewing hackathon documentation alignment and checking word-count constraints	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-assisted code, UI styling, and documentation drafts were treated as untrusted proposals subject to human engineering review. Our tech lead and team manually audited all scoring formulas, keyword-to-skill mapping rules, and API contracts to guarantee deterministic, explainable outputs and eliminate fabricated courses, statistics, or black-box LLM dependencies at runtime. We verified end-to-end functionality through automated and manual browser/API capture tests (output/pitch/demo-proof.json), explicitly testing scenario transitions, state persistence across reloads, revert actions, and strict disclosure of planning assumptions.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud Development & Deployment Platform (`replit.app`)	Used to build, containerize, and deploy the live full-stack Nexus web application (serving both the React/Vite frontend workspace and Node.js/Express Career API at `https://NexusTalent-Decisive.replit.app`).

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the cloud compute and public web hosting for our live full-stack prototype (https://NexusTalent-Decisive.replit.app), serving both the compiled React/Vite frontend and the Node.js/Express Career API from a single unified deployment. While Replit streamlines live judge access and instant cloud deployment, our application uses standard Node.js/Express and Vite without proprietary lock-in, meaning it can also be run locally or migrated to any standard Node.js cloud host without altering core functionality.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

In the individual worker workspace, uploaded CV files are read in the browser (PDF.js/fflate); extracted text and reviewed profiles are processed transiently in memory by the Express API without server persistence, while saved decisions, completions, and notes remain exclusively in device localStorage .
In the **HR & Employers workflow (/hr/demo and /survey/:token)**, the public prototype enforces a strict **"Demo Mode: Use synthetic data only — do not enter real personal data or real CVs"** policy. Seeded departments (24 synthetic profiles) and expiring /survey/:token campaign submissions are saved to a local, single-process JSON store on the server rather than a production database. No personal data is ever transmitted to third-party services or AI providers; outbound requests to the public Crossref REST API contain only fixed research topic queries .

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1.
2. **Proxy Signal & Forecast Limits: Crossref keyword mappings and scenario multipliers (*Steady, Faster AI, Green transition*) are bounded planning assumptions, not Malaysian vacancy forecasts, HRD Corp claimability guarantees, or measured SDG outcomes. Mitigation: The UI separates retrieved source receipts from Nexus mapping assumptions and labels all scores as discussion indices .**
3. **External API Rate Limits / Downtime: Live Crossref refreshes may fail. Mitigation: The backend maintains a versioned disk revision cache with explicit *"Stale cache"* status badges and neutral offline fallback weights .**

4. **Rule-Based Matching & Public HR Demo Constraints: Deterministic keyword matching does not infer proficiency, and the public HR demo (/hr/demo, /survey/:token) uses a single-process local JSON store without employer authentication. Mitigation: HR links are quota-limited and revocable, synthetic-only data warnings are enforced, and all briefs are exported for human review .**

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]
24/10/26